



Start Here

Start from the engineering intent, visually compare the governed authorities, then open only the reference you need.

[Search](#)[Theme · System](#)[Appearance](#)

NICEGUI BASE · DEPARTMENT STANDARD

What are you building?

Start from intent. NiceGUI Base will lead you to the governed pattern, layout, visualization, component, recipe and scaffold instead of making you browse hundreds of references one by one.

10

Patterns

58

Analytics

34

Components

8

Recipes

Choose an intent

See the recommended pattern, recipe, scaffold, and related governed references before leaving this page.

Developer Scaffolding

This saved-link route now points developers to deterministic pattern and recipe starters.

Search



Theme · System

Appearance

Compatibility authoring surface

Manual composition is retired from the normal reference journey. Start from a canonical pattern or recipe with the installed CLI; domain logic stays in your application services.

CLI · CREATE-PATTERN

Choose a pattern

Use a registered page hierarchy and responsive slot contract.

CLI · CREATE-RECIPE

Choose a recipe

Use a bounded semiconductor workflow and its data contract.

AGENT WORKFLOW

Read the agent path

Resolve a requirement into an authority before coding.



Design System

The deterministic visual contract used by the reference explorer, framework components, and generated applications.

[Search](#)[Theme · System](#)[Appearance](#)

Design System Authority

See the token in use first; the compact contract beside each specimen is the source for Python consumers, components, CSS, and generated applications.

Import from `nicegui_base.design`; choose semantic roles before extending the framework.

Spacing scale

Use for recurring distances; prefer semantic gaps for composition.

LIVE USE

**Tight**`--cui-space-1 · 4px`**Related**`--cui-space-2 · 8px`**Stack**`--cui-space-4 · 16px`**Content**`--cui-space-6 · 24px`**Section**`--cui-space-10 · 40px`

Settings

Manage appearance and Explorer identity without leaving the department reference shell.

[Search](#)[Theme · System](#)[Appearance](#)

Appearance

These preferences apply to the Explorer and its mounted reference previews. System follows the operating-system color scheme.

Theme[System](#)[Light](#)[Dark](#)**Density**[Comfort](#)[Compact](#)[Dense](#)**Motion**[Normal](#)[Reduced](#)

Account

Identity is read from the company environment and is display-only here.

AccessKey

Company identity value

Design System

One discoverable inventory over canonical NiceGUI Base registries, grouped by the job each capability solves.

Search



Theme · System

Appearance

Search catalog

Name, tag, intent, data shape, dc

Refine results · type, family, intent, data, domain, relation

Analytics Studio

465 canonical Reference Explorer entries are searchable.
Choose a family or search to inspect individual capabilities.

Shell/Layout

1 canonical entry

Examples: Layout

Browse family

Controls

35 canonical entries

Examples: ButtonGroup, SplitButton, Divider

Components

Scan governed component specimens first; open a detail only for deeper states, configuration, accessibility, or code.

Search



Theme · System

Appearance

ButtonGroup

interactive specimen Use the real control below; interactions are recorded in @event / debug output.

View Pass Reset

ACTIONS · LIVE

ButtonGroup

Group related actions without visual fragmentation

Best for · toolbar actions · mode actions

→ Open example

☆ Favorite

Compare

SplitButton

interactive specimen Use the real control below; interactions are recorded in @event / debug output.

⌵ Export File

Application Patterns

Choose a finished application structure, see the real governed composition, then adapt business logic in your own services.

Search

⚙ Theme · System

Appearance

LIVE GOVERNED COMPOSITION

Dashboard

kpis · trend · exceptions

Dashboard

At-a-glance process performance with trends and exceptions.

Lots today

128

↑ +6%

Across three qualified tools

In control

96.9%

↑ +0.8 pp

Open exceptions

3



Visualizations

See what all 58 governed analytical surfaces actually look like before opening a full interactive reference.

Search



Theme · System

Appearance

Individuals — SPC I-MR



Show chart data

Moving Range — SPC I-MR



SPC · LIVE

SPC I-MR

Individuals and moving-range control chart

Best for · continuous measurements without rational subgroups

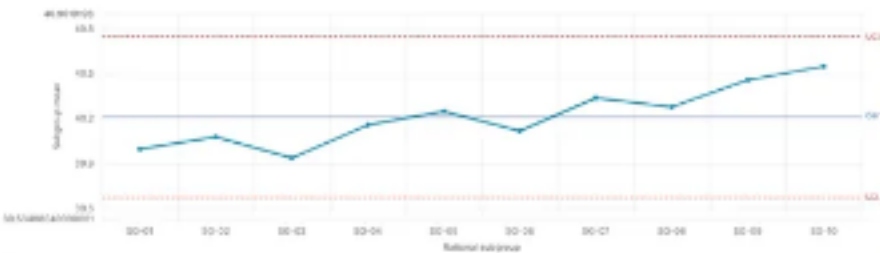


Open example



Favorite

Compare



Show chart data

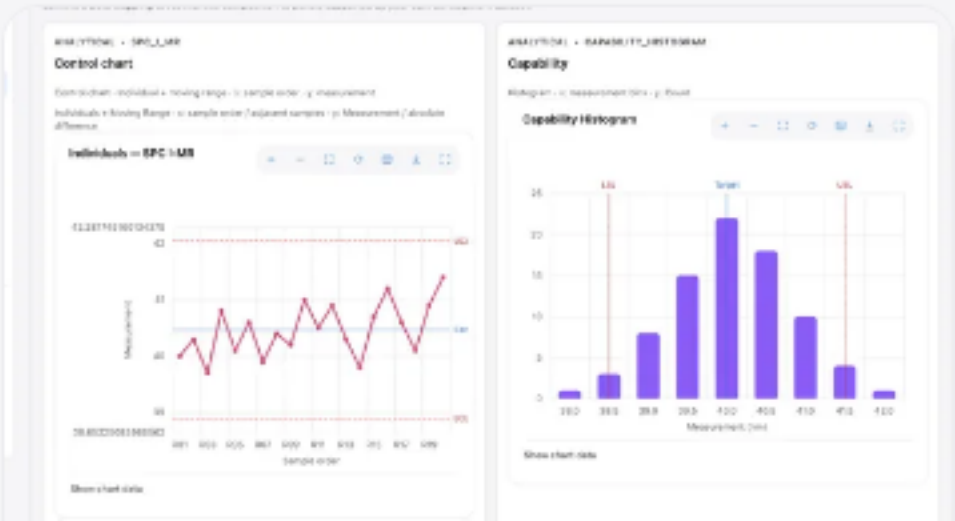
Semiconductor Recipes

Scan the domain workflows visually, compare their contracts, then open the recipe that matches the engineering question.

Search

 Theme · System

Appearance



SEMICONDUCTOR RECIPE · LIVE

SPCMonitor

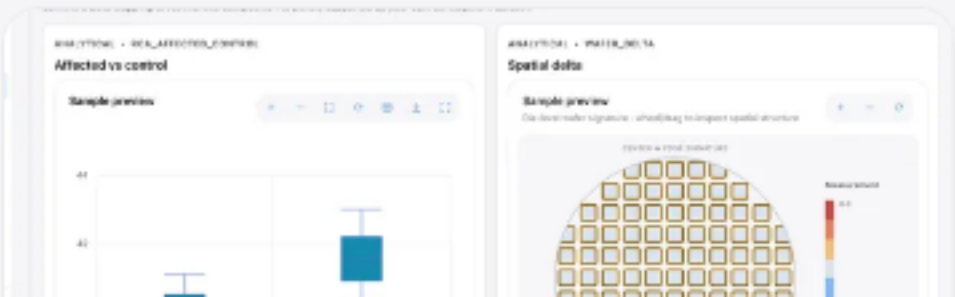
Monitor process stability, capability and violations with drill-down to manufacturing context.

Best for · monitor a process parameter · detect process drift or control-rule violations

→ Open example

☆ Favorite

Compare



Data & Tables

A governed production reference for tables, record management, data exploration and provider-scale grids.

Search



Theme · System

Appearance

Production table lab

The governed production reference for tables, record management, data exploration and provider-scale grids. One active lab keeps the page fast; every lab reuses the same semiconductor data contract.

Grid

Actions

Edit

Visualize

Import

Detail

Grid

The production read-oriented grid: query, layout, views and persisted state.

64 records

Persisted view



Watch / OOS



Unreviewed

Diagnostics

Understand runtime health, framework coverage, and the few actions that need attention.

Search

 Theme · System

Appearance

System health

A quick answer to whether the installed reference is ready to use.

Reference Explorer

Healthy

Candidate NGB-20260907-G2.6 · NiceGUI Base 3.0.0a8
· NiceGUI 3.15.0

Generic application patterns

Available	10
Discoverable	10 / 10
Live examples	10 / 10

Semiconductor analytics

Available	58
Discoverable	58 / 58
Live examples	58 / 58

AI Development Guide

A compact, task-first guide for humans and AI agents using the installed NiceGUI Base APIs.

Search



Theme · System

Appearance

Requirement → runnable reference

The same deterministic sequence is readable by a person and discoverable by an agent.

1 · REQUIREMENT

Investigate chamber drift and compare affected/control distributions

2 · RECOMMENDATION

Pattern: analysis_workspace · Recipe: excursion-defense-line

3 · AUTHORITIES

analytics:fdc_multi_sensor ·
analytics:rca_affected_control ·
component:data_table

4 · SCAFFOLD



Foundation

Inspect the exact visual primitives from which every application is constructed.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Semantic color system

These are design roles, not a decorative palette. Apps should never invent independent visual colors.

Page
--cui-page

Primary surface
--cui-surface

Secondary surface
--cui-surface-secondary

Elevated
--cui-surface-elevated

Shell Primitives

One responsive navigation state machine: expanded desktop, icon-only desktop, or temporary mobile overlay.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Responsive navigation contract

Desktop has one collapse control inside the rail. Below 900px the rail is removed from interaction and one navigation trigger appears with the header actions — never beside the application title.

Expanded desktop

Icon-only desktop

Mobile overlay

Resize the browser to exercise the states. There is intentionally no standalone mobile-navigation demo action because that created a competing navigation system.

Header interaction contract

Application settings and user profile are real structured popovers. Settings navigates to the canonical settings workspace; the user control



Control Laboratory

Every primitive shown against normal, disabled, selected, error, long-content and compact conditions.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Buttons and action hierarchy

Intent hierarchy

Primary

Secondary

Tertiary

Ghost

Danger

Save changes

Processing

Disabled

Icon, group and split actions



Forms, Overlays & Feedback

Exercise real form anatomy, dirty state, dialogs, drawers, menus, progress, alerts and failure recovery.

THEME

SystemLightDark

DENSITY

ComfortCompactDense

MOTION

NormalReduced

Resize the real browser window for responsive proof

Production form

Routing

Changes here are intentionally marked dirty for navigation protection.

Preferred tool

ETCH-021

×

▼

Recipe *

ETCH_R18



Automatic dispatch

Cancel





Enterprise DataTable Gallery

Five distinct production patterns prove different table jobs instead of repeating one measurement dataset.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

1 • Measurement population

High-volume read-only analysis: search, sort, filter, resize, select, conditional values, status, sparklines, bulk actions and row inspection.

Measurement population

320 deterministic records · dense analytical browsing · double-click any row for inspection

Search records

☐	Measurement	Actions
☐	MEAS-100000	... Actions
☐	MEAS-100001	... Actions
☐	MEAS-100002	... Actions



Visualization Recipe Gallery

Compact reusable recipes demonstrate analytical intent, markings, statistics, interactions and specialist visual forms —not just renderer types.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

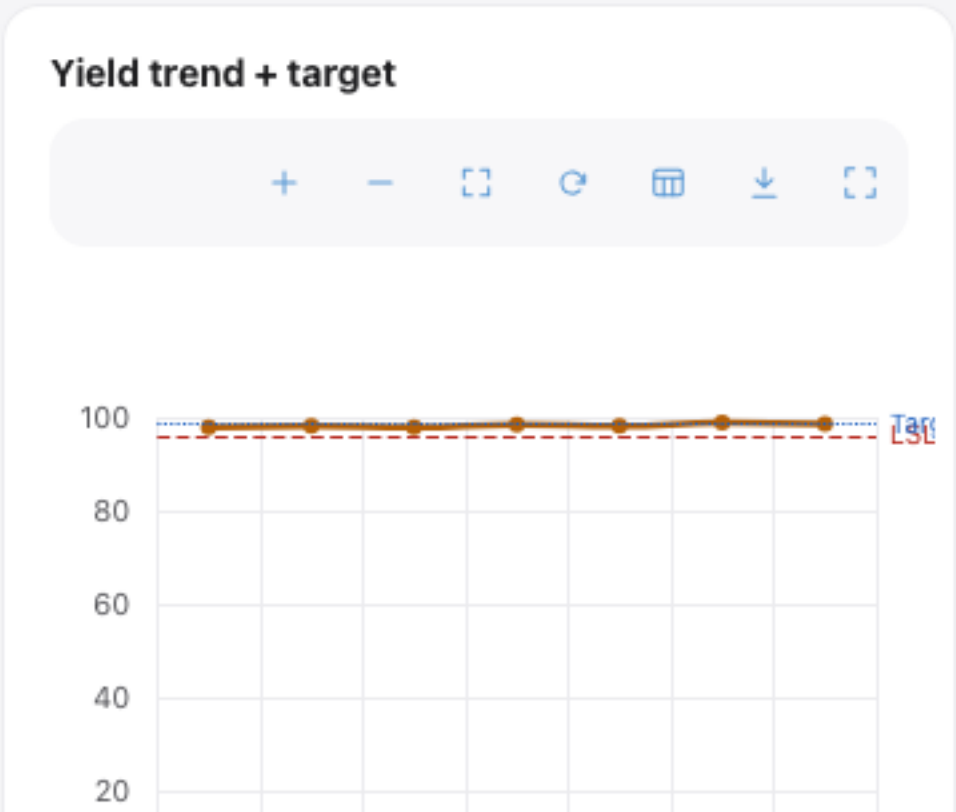
Normal

Reduced

Resize the real browser window for responsive proof

Trend & time recipes

Premade temporal patterns include limits, target context, multi-series comparison, events, confidence bands and step changes.





Content, Workflow & Commands

Viewers and workflow components should visually inherit NiceGUI Base rather than their underlying third-party defaults.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Entity and property presentation

ETCH-021 / CH-3

Watch

Critical etch chamber investigation

Chamber Area: ETCH Recipe: ETCH_R18

Key/value
Lot
L260142
Wafer
W12
Owner
Process Engineering



Engineering & RCA Laboratory

Purpose-built investigation cockpit: scope, process behavior, population separation, evidence, commonality, contradictions and timeline.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

⌘ Affected population

84 wafers

ⓘ 12 lots

⊕ Matched controls

215 wafers

ⓘ exact route strategy

📄 Evidence confidence

High

ⓘ 2 strong channels



State & Failure Laboratory

Every unhappy path must still look intentional and preserve clear recovery behavior.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Canonical application states



No data yet

Connect a source or adjust the time window.

Configure



No matching results

Filters excluded all 18,402 measurements.



Performance Laboratory

Interactive workloads expose generation, render assembly, cancellation and query timing instead of hiding work behind a button.

THEME

System

Light

Dark

DENSITY

Comfort

Compact

Dense

MOTION

Normal

Reduced

Resize the real browser window for responsive proof

Interactive workload benchmark

Every action updates visible state immediately. The large workload remains opt-in so ordinary lab navigation stays fast.

Local grid workload

Generate deterministic synthetic rows off the UI event loop, mount the real Company DataTable, then expose timing and status.

Idle

0

Rows

—

Generation

—

Table mount

—

Filter benchmark



REFERENCE APP PREVIEW • DASHBOARD

Dashboard

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Yield Performance

Executive-to-engineering overview with one clear reading order

Area

All areas



24 h

7 d

30 d



Yield

98.4%

↑ +0.7% WoW



Excursions

12

↑ 3 critical



Fleet health

94.6%



REFERENCE APP PREVIEW • DATA EXPLORER

Data Explorer

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Process Measurement Explorer

Filter → summarize → analyze → inspect without losing population context

Area

Tool

All areas x v

All tools x v

24 h

7 d

30 d

Population

18,402

i selected rows

Mean CD

40.62 nm

i +0.31 nm

REFERENCE APP PREVIEW • MASTER DETAIL

Master Detail

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Equipment Fleet

Browse a dense fleet while selected-equipment context stays visible

Area

Tool

All areas

x

▼

All tools

x

▼

Equipment signals

Q Search records			   	
Measurement			Lot	
☐	MEAS-100000		L260100	
☐	MEAS-100001		L260100	
☐	MEAS-100002		L260100	
☐	MEAS-100003		L260100	

REFERENCE APP PREVIEW • CRUD

Crud

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Saved Investigation Views

Create, inspect and manage reusable engineering configurations

Search saved views

All Mine Shared

+ New saved view

Saved views	
Search records	
   	
ID	Name



REFERENCE APP PREVIEW • MONITORING

Monitoring

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Fleet Monitoring

Operational health leads; investigation context stays one click away

Area

All areas x v 24 h 7 d 30 d

Healthy

84

Watch

7

Critical

3



REFERENCE APP PREVIEW • SEARCH

Search

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Engineering Search

Search-first navigation for tools, chambers, investigations and evidence

Search everything

ETCH-021



Refine

- ☒ Equipment
- ☒ Investigations
- ☒ Evidence

3 results • indexed 2 min ago

ETCH-021
Tool • Watch



REFERENCE APP PREVIEW • WIZARD

Wizard

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

[Back to Reference Explorer](#)

Create Investigation

Guided task with visible progress, bounded decisions and a real review step



Scope
Complete



Controls
Active



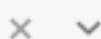
Rev
Upco

Select control population

Use the narrowest defensible comparison population before reviewing evidence.

Control strategy

Exact route match



Require exact tool / recipe / chamber sequence



Exclude wafers processed after maintenance

REFERENCE APP PREVIEW • COMPARISON

Comparison

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Affected vs Control

Direct comparison keeps population identity, deltas and spatial evidence aligned

Affected population

EXC-1042 · 84 wafers × ▾

Control population

Matched route · 215 wafers × ▾

Affected mean

42.18 nm

Control mean

39.96 nm

Delta



REFERENCE APP PREVIEW • ANALYSIS WORKSPACE

Analysis Workspace

Interactive canonical pattern. Explore the real behavior here, then use the same registered pattern from the developer scaffolding CLI.

Back to Reference Explorer

Excursion Investigation — EXC-1042

Dense analytical workspace with persistent orientation and semiconductor-native evidence

INVESTIGATION

EXC-1042 · Chamber 3 process degradation

OWNER

Process Engineering

STAGE

Evidence review

UPDATED

2 minutes ago

Affected

84 wafers

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

COMPONENT · ACTIONS

ButtonGroup

Group related actions without visual fragmentation

BEST FOR

toolbar actions · mode actions

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · ACTIONS

SplitButton

Primary action with closely related alternatives

BEST FOR

export alternatives · run options

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Inspect

Code

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · SURFACES

Divider

Subtle structural separation

BEST FOR

sections · menus

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)[COMPONENT](#) · [SURFACES](#)

CollapsiblePanel

Progressive disclosure without leaving context

BEST FOR

advanced settings · secondary detail

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

[Preview](#)[Usage](#)[Configure](#)[States](#)[Interactions](#)

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · SURFACES

Accordion

Grouped progressive disclosure

BEST FOR

help · configuration groups

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interactions

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · STATUS

Chip

Interactive compact metadata or filter value

BEST FOR

filter value · selection

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

**Advanced contract ·
> configuration, data,
accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · STATUS

CountBadge

Compact numeric count

BEST FOR

notifications · selected count

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Inspect

Code

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · STATUS

SeverityIndicator

Semantic severity with label and non-color cue

BEST FOR

operational state · risk

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · STATUS

FreshnessIndicator

Data recency state

BEST FOR

updated time · stale data

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · STATUS

DataQualityBadge

Data reliability/completeness state

BEST FOR

partial data · estimated data

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.



Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

COMPONENT · ACTIONS

Button

Standard user action

BEST FOR

normal action · secondary action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

[Preview](#)[Usage](#)[Configure](#)[States](#)[Interactions](#)

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · ACTIONS

ActionButton

Async-safe or stateful action

BEST FOR

save · run analysis

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interact

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · ACTIONS

IconButton

Compact icon-only action with accessible label

BEST FOR

toolbar action · row action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · SURFACES

Surface

Consistent content containment and interaction surface

BEST FOR
panel · card

AVOID WHEN
Do not substitute an unregistered visual or state implementation.

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · STATUS

Badge

Compact semantic status or metadata label

BEST FOR
status · severity

AVOID WHEN
Do not substitute an unregistered visual or state implementation.

Advanced contract ·
configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · INPUTS

TextInput

Single-line text entry

BEST FOR

name · identifier

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · INPUTS

NumberInput

Numeric entry with bounds and unit

BEST FOR

threshold · count

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · INPUTS

TextArea

Multi-line text entry

BEST FOR

notes · description

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · INPUTS

SearchInput

Debounced search entry

BEST FOR

global search · table search

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · INPUTS

Select

Single choice from known values

BEST FOR

filter · form choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interact

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

COMPONENT · INPUTS

MultiSelect

Multiple choices from known values

BEST FOR

multi-filter · assignment

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

[Preview](#)[Usage](#)[Configure](#)[States](#)[Interactions](#)

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · INPUTS

Autocomplete

Searchable known values

BEST FOR

large option set · entity lookup

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Configure

States

Interact

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

COMPONENT · INPUTS

Combobox

Search/select with optional custom value

BEST FOR

tag-like entry · mixed known/custom choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

[Preview](#)[Usage](#)[Configure](#)[States](#)[Interactions](#)

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

COMPONENT · INPUTS

Checkbox

Independent boolean selection

BEST FOR

multi-option form · enable choice

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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COMPONENT · INPUTS

CheckboxGroup

Set of independent boolean choices

BEST FOR

feature selection · permissions

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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COMPONENT · INPUTS

RadioGroup

Mutually exclusive choice

BEST FOR

mode selection · single preference

AVOID WHEN

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> configuration, data,
accessibility, and authority

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COMPONENT · INPUTS

Switch

Immediate on/off setting

BEST FOR

preference toggle · feature enablement

AVOID WHEN

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Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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COMPONENT · INPUTS

Slider

Bounded continuous or stepped value

BEST FOR

threshold · range tuning

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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COMPONENT · INPUTS

RangeSlider

Bounded low/high selection

BEST FOR

numeric filtering · window selection

AVOID WHEN

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Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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COMPONENT · INPUTS

DatePicker

Single date selection

BEST FOR

effective date · event date

AVOID WHEN

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Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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Theme · System

Appearance

COMPONENT · INPUTS

DateRangePicker

Date interval selection

BEST FOR

analysis period · reporting window

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Preview

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Interact

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

COMPONENT · INPUTS

TimePicker

Time-of-day selection

BEST FOR

schedule time · cutoff

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

COMPONENT · INPUTS

DateTimePicker

Date and time selection

BEST FOR

timestamp · scheduled action

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

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> configuration, data,
accessibility, and authority

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Reference Explorer

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Search



Theme · System

Appearance

COMPONENT · INPUTS

FileUpload

Validated file selection and upload

BEST FOR

attachment · data import

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Preview

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Dashboard

High-level KPI and trend overview.

BEST FOR

High-level KPI and trend overview.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Live example is ready.

Preview

Usage

Data

Inspect

Code

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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PATTERN · APPLICATION PATTERN

Data Explorer

Interactive filtering, analysis, records and contextual drill-down.

BEST FOR

Interactive filtering, analysis, records and contextual drill-down.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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PATTERN · APPLICATION PATTERN

Master Detail

Browse entities while preserving selected-entity context.

BEST FOR

Browse entities while preserving selected-entity context.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Crud

Search, create, inspect and edit managed records.

BEST FOR

Search, create, inspect and edit managed records.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

PATTERN · APPLICATION PATTERN

Monitoring

Operational health, alerts and periodically refreshed data.

BEST FOR

Operational health, alerts and periodically refreshed data.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

PATTERN · APPLICATION PATTERN

Search

Search and refine heterogeneous or entity-oriented results.

BEST FOR

Search and refine heterogeneous or entity-oriented results.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Settings

Structured application/user configuration.

BEST FOR

Structured application/user configuration.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

**Advanced contract ·
> configuration, data,
accessibility, and authority**

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Preview

Usage

Inspect

Code



Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)

PATTERN · APPLICATION PATTERN

Wizard

Guided multi-step task with clear progress and bounded decisions.

BEST FOR

Guided multi-step task with clear progress and bounded decisions.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

PATTERN · APPLICATION PATTERN

Comparison

Compare baseline/current populations, entities or scenarios.

BEST FOR

Compare baseline/current populations, entities or scenarios.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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PATTERN · APPLICATION PATTERN

Analysis Workspace

Maximum-density resizable chart/table analysis environment.

BEST FOR

Maximum-density resizable chart/table analysis environment.

AVOID WHEN

Do not substitute an unregistered visual or state implementation.

Advanced contract ·
> configuration, data,
accessibility, and authority

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · SPC

SPC I-MR

Individuals and moving-range control chart

BEST FOR

continuous measurements without rational subgroups

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

Related capabilities

[SPCMonitor](#)

[PMEffectAnalysis](#)

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

[Search](#)[Theme · System](#)[Appearance](#)[ANALYTIC](#) · [SPC](#)

SPC Xbar-R

Xbar-R subgroup control chart

BEST FOR

small equal-size rational subgroups

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · SPC

SPC Xbar-S

Xbar-S subgroup control chart

BEST FOR

equal-size subgroup variability matters

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Preview

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Reference Explorer

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ANALYTIC · SPC

SPC P

Proportion nonconforming control chart

BEST FOR

binary defect proportion with varying samples

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

ANALYTIC · SPC

SPC Np

Count nonconforming control chart

BEST FOR

binary defect counts with constant sample size

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · SPC

SPC C

Defect count control chart

BEST FOR

Poisson defect counts with constant opportunity

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · SPC

SPC U

Defects-per-unit control chart

BEST FOR

Poisson defects with varying opportunity

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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SPC EWMA

EWMA drift-sensitive control chart

BEST FOR

small sustained process shifts

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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ANALYTIC · SPC

SPC CUSUM

CUSUM change-detection chart

BEST FOR

small persistent mean shifts

AVOID WHEN

when chart assumptions, subgroup structure, or baseline definition do not match the data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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ANALYTIC · CAPABILITY

Capability Histogram

Capability distribution with spec limits

BEST FOR

Cp/Cpk/Pp/Ppk and distribution shape

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

[SPCMonitor](#)

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Reference Explorer

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Search



Theme · System

Appearance

ANALYTIC · CAPABILITY

Q-Q Probability

Normal probability/QQ diagnostic

BEST FOR

capability normality assumption review

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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Theme · System

Appearance

ANALYTIC · CAPABILITY

ECDF

Empirical cumulative distribution

BEST FOR

distribution comparison without binning

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

SPCMonitor

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Reference Explorer

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Appearance

ANALYTIC · CAPABILITY

Box Distribution

Box distribution comparison

BEST FOR

compact affected/control distribution summaries

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract ·
➤ configuration, data,
accessibility, and authority

Related capabilities

PMEffectAnalysis

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Live example is ready.

Reference Explorer

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ANALYTIC · CAPABILITY

Violin Distribution

Violin distribution comparison

BEST FOR

shape plus spread comparison

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract ·
> configuration, data,
accessibility, and authority

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Appearance

ANALYTIC · CAPABILITY

Ridge Distribution

Ridge distribution comparison

BEST FOR

many tool/chamber distributions

AVOID WHEN

when specification limits, measurement-system quality, or population definition are unknown

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

ChamberMatching

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Live example is ready.

Reference Explorer

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Search



Theme · System

Appearance

ANALYTIC · WAFER

Wafer Continuous

Continuous wafer map

BEST FOR

continuous spatial metrology or process response

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
➤ configuration, data,
accessibility, and authority

Related capabilities

[SPCMonitor](#)

[LotWaferExplorer](#)

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Categorical

Categorical/bin wafer map

BEST FOR

yield bins or categorical spatial state

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

- Advanced contract ·**
➤ **configuration, data, accessibility, and authority**

Related capabilities

[LotWaferExplorer](#)

[YieldLoss](#)

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Defect

Defect wafer map

BEST FOR

inspection defects and defect classes

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
> configuration, data,
accessibility, and authority

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Delta

Affected minus control wafer delta map

BEST FOR

spatial process change localization

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
> configuration, data,
accessibility, and authority

Related capabilities

[ExcursionDefenseLine](#)

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Comparison

Side-by-side wafer comparison

BEST FOR

affected/control wafer signature comparison

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
➤ configuration, data,
accessibility, and authority

Related capabilities

ChamberMatching

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Live example is ready.

Reference Explorer

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Search



Theme · System

Appearance

ANALYTIC · WAFER

Lot Wafer Strip

Lot wafer strip

BEST FOR

lot-level wafer-to-wafer spatial variation

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
› configuration, data,
accessibility, and authority

Related capabilities

LotWaferExplorer

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Small Multiples

Synchronized wafer small multiples

BEST FOR

many wafers on one scale

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
> configuration, data,
accessibility, and authority

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Contour

Contour/surface wafer map

BEST FOR

data-derived spatial field interpretation

AVOID WHEN

sparse categorical data

Advanced contract ·
> configuration, data,
accessibility, and authority

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Radial

Radial profile

BEST FOR

center-edge signatures

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
› configuration, data,
accessibility, and authority

Related capabilities

LotWaferExplorer

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Reference Explorer

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Center Edge

Center-edge decomposition

BEST FOR

quantify center versus edge shift

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
> configuration, data,
accessibility, and authority

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Reference Explorer

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Ring

Ring analysis

BEST FOR

concentric process signatures

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
> configuration, data,
accessibility, and authority

Related capabilities

[LotWaferExplorer](#)

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Sector

Sector analysis

BEST FOR

azimuthal spatial signatures

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
➤ configuration, data,
accessibility, and authority

Related capabilities

LotWaferExplorer

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Theme · System

Appearance

ANALYTIC · WAFER

Wafer Defect Clusters

Defect-cluster overlays

BEST FOR

identify spatially clustered inspection defects

AVOID WHEN

when coordinates are unavailable or spatial sampling is too sparse for the intended inference

Advanced contract ·
› configuration, data,
accessibility, and authority

Related capabilities

LotWaferExplorer

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · FDC

FDC Recipe Step Trace

Recipe-step-aligned trace

BEST FOR

compare traces with step semantics

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

FdcToolHealth

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · FDC

FDC Golden Envelope

Golden trace envelope

BEST FOR

compare trace deviation against baseline population

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
➤ configuration, data,
accessibility, and authority

Related capabilities

FdcToolHealth

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · FDC

FDC Multi Sensor

Multi-sensor trace panel

BEST FOR

correlated equipment sensor behavior

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

- Advanced contract ·**
- **configuration, data, accessibility, and authority**

Related capabilities

FdcToolHealth

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

Search



Theme · System

Appearance

ANALYTIC · FDC

FDC Tool Chamber Compare

Tool/chamber comparison

BEST FOR

matching and PM-effect comparisons

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

[PMEffectAnalysis](#)

[ChamberMatching](#)

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.



Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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ANALYTIC · FDC

FDC Chamber Fingerprint

Chamber fingerprint

BEST FOR

multivariate chamber matching

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
> configuration, data,
accessibility, and authority

Related capabilities

[FdcToolHealth](#)[PMEffectAnalysis](#)[ChamberMatching](#)

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Live example is ready.

Reference Explorer

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Theme · System

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ANALYTIC · FDC

FDC Sensor Fingerprint

Sensor fingerprint

BEST FOR

sensor signature comparison

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
> configuration, data, accessibility, and authority

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ANALYTIC · FDC

FDC Alarm Overlay

Alarm overlay

BEST FOR

align alarms with trace excursions

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
> configuration, data,
accessibility, and authority

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ANALYTIC · FDC

FDC Equipment Event Overlay

Equipment event overlay

BEST FOR

align PM/recipe/equipment events with traces

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

[FdcToolHealth](#)[PMEffectAnalysis](#)

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ANALYTIC · FDC

FDC PCA Scores

PCA scores

BEST FOR

multivariate operating-state separation

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

FdcToolHealth

ChamberMatching

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ANALYTIC · FDC

FDC PCA Loadings

PCA loadings

BEST FOR

variables driving multivariate separation

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

FdcToolHealth

ChamberMatching

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ANALYTIC · FDC

FDC Hotelling T²

Hotelling T²

BEST FOR

multivariate score-space anomaly magnitude

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
> configuration, data, accessibility, and authority

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ANALYTIC · FDC

FDC SPE Q

SPE/Q statistic

BEST FOR

PCA residual anomaly magnitude

AVOID WHEN

when sensor semantics, recipe-step alignment, or event timing are unavailable

Advanced contract ·
> configuration, data,
accessibility, and authority

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Theme · System

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ANALYTIC · RCA

RCA Affected Control

Affected/control population comparison

BEST FOR

quantify excursion shift without causal overclaim

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
› configuration, data,
accessibility, and authority

Related capabilities

[ExcursionDefenseLine](#)

[PMEffectAnalysis](#)

[RcaCockpit](#)

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ANALYTIC · RCA

RCA Commonality Ranking

Commonality ranking

BEST FOR

rank shared factors across affected versus control

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
› configuration, data,
accessibility, and authority

Related capabilities

ExcursionDefenseLine

YieldLoss

RcaCockpit

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ANALYTIC · RCA

RCA Enrichment

Enrichment analysis

BEST FOR

factor exposure over-representation

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
> configuration, data,
accessibility, and authority

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ANALYTIC · RCA

RCA Commonality Matrix

Commonality matrix

BEST FOR

scan factor/value enrichment

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

[ExcursionDefenseLine](#)

[RcaCockpit](#)

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ANALYTIC · RCA

RCA Contribution Waterfall

Contribution waterfall

BEST FOR

show ranked contribution accumulation

AVOID WHEN

as proof of causality without corroborating engineering evidence

- Advanced contract ·**
- **configuration, data, accessibility, and authority**

Related capabilities

RcaCockpit

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ANALYTIC · RCA

RCA Correlation Matrix

Correlation matrix

BEST FOR

numeric relationship screening

AVOID WHEN

causal conclusions

Advanced contract ·
> configuration, data,
accessibility, and authority

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Preview

Usage

Data

Inspect

Code

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ANALYTIC · RCA

RCA Evidence Matrix

Evidence matrix

BEST FOR

hypothesis evidence/contradiction review

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

[ExcursionDefenseLine](#)

[RcaCockpit](#)

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ANALYTIC · RCA

RCA Genealogy Graph

Genealogy/process graph

BEST FOR

lot/wafer/tool/recipe route genealogy

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

[ExcursionDefenseLine](#)

[RcaCockpit](#)

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Theme · System

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ANALYTIC · RCA

RCA Cause Tree

Cause tree

BEST FOR

structured causal candidates

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

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Preview

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ANALYTIC · RCA

RCA Fault Tree

Fault tree

BEST FOR

logical failure decomposition

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

RcaCockpit

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ANALYTIC · RCA

RCA Sankey

Sankey/process flow

BEST FOR

population flow through route/tool/process paths

AVOID WHEN

as proof of causality without corroborating engineering evidence

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

RcaCockpit

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Theme · System

Appearance

ANALYTIC · YIELD

Yield Pareto

Yield loss Pareto

BEST FOR

rank yield-loss contributors

AVOID WHEN

when numerator/denominator or bin definitions are inconsistent across populations

Advanced contract ·
➤ **configuration, data, accessibility, and authority**

Related capabilities

YieldLoss

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Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

Appearance

ANALYTIC · YIELD

Bin Pareto

Bin Pareto

BEST FOR
rank failing bins

AVOID WHEN
when numerator/denominator or bin definitions
are inconsistent across populations

Advanced contract ·
➤ **configuration, data,
accessibility, and authority**

Related capabilities

YieldLoss

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and never modifies a project.

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Reference Explorer

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Theme · System

Appearance

ANALYTIC · YIELD

Yield Waterfall

Yield waterfall/decomposition

BEST FOR

explain yield delta composition

AVOID WHEN

when numerator/denominator or bin definitions are inconsistent across populations

- Advanced contract ·**
- configuration, data, accessibility, and authority

Related capabilities

YieldLoss

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Live example is ready.

Reference Explorer

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Theme · System

Appearance

ANALYTIC · RELIABILITY

Weibull Reliability

Weibull reliability analysis

BEST FOR

time-to-failure reliability data

AVOID WHEN

when failure/censoring semantics or exposure time are unknown

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

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ANALYTIC · DOE

DOE Main Effects

DOE main effects

BEST FOR

screen factor effects

AVOID WHEN

for observational data that were not generated by a valid designed experiment

Advanced contract ·
> configuration, data,
accessibility, and authority

Reference example · governed sample data stays on this page and never modifies a project.

Live example is ready.

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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Theme · System

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ANALYTIC · DOE

DOE Interactions

DOE interaction plot

BEST FOR

factor interaction assessment

AVOID WHEN

for observational data that were not generated by a valid designed experiment

Advanced contract ·
> configuration, data,
accessibility, and authority

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ANALYTIC · DOE

DOE Response Surface

DOE response surface

BEST FOR

quadratic two-factor response optimization

AVOID WHEN

for observational data that were not generated by a valid designed experiment

Advanced contract ·
> configuration, data,
accessibility, and authority

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RECIPE · SEMICONDUCTOR RECIPE

SPCMonitor

Monitor process stability, capability and violations with drill-down to manufacturing context.

BEST FOR

monitor a process parameter · detect process drift or control-rule violations

AVOID WHEN

when the process has no defensible baseline or chart family/subgroup definition

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[SPC I-MR](#)[Capability Histogram](#)[ECDF](#)[Wafer Continuous](#)

Reference Explorer

Browse the canonical capability, try its live example, and inspect the reusable usage contract.

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RECIPE · SEMICONDUCTOR RECIPE

ExcursionDefenseLine

Triage an excursion with affected/control populations, spatial localization, commonality and genealogy evidence.

BEST FOR

build an excursion defense line · triage affected versus control material

AVOID WHEN

when affected/control population definitions are not yet governed or comparable

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[RCA Affected Control](#)[Wafer Delta](#)[RCA Commonality Ranking](#)[RCA Commonality Matrix](#)[RCA Genealogy Graph](#)[RCA Evidence Matrix](#)

Reference Explorer

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RECIPE · SEMICONDUCTOR RECIPE

FdcToolHealth

Diagnose equipment health from aligned sensor traces, envelopes, events and multivariate fingerprints.

BEST FOR

analyze FDC traces · check tool or chamber health

AVOID WHEN

when recipe-step alignment or sensor/event semantics are unavailable

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[FDC Recipe Step Trace](#)[FDC Golden Envelope](#)[FDC Multi Sensor](#)[FDC Chamber Fingerprint](#)[FDC PCA Scores](#)[FDC PCA Loadings](#)[FDC Equipment Event Overlay](#)

Reference Explorer

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RECIPE · SEMICONDUCTOR RECIPE

LotWaferExplorer

Explore lot-to-wafer-to-die spatial behavior with synchronized maps, defects, profiles and records.

BEST FOR

explore wafers in a lot · inspect spatial metrology or defect signatures

AVOID WHEN

when wafer/die coordinates or wafer identity are unavailable

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[Wafer Continuous](#)[Wafer Categorical](#)[Lot Wafer Strip](#)[Wafer Radial](#)[Wafer Ring](#)[Wafer Sector](#)[Wafer Defect Clusters](#)



Reference Explorer

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RECIPE · SEMICONDUCTOR RECIPE

YieldLoss

Explain yield loss through Pareto/decomposition, bin/spatial signatures and commonality drill-down.

BEST FOR

analyze yield loss · rank fail bins or loss contributors

AVOID WHEN

when yield denominators, bin definitions or comparison populations are inconsistent

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[Yield Pareto](#)[Bin Pareto](#)[Yield Waterfall](#)[Wafer Categorical](#)[RCA Commonality Ranking](#)

Reference Explorer

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RECIPE · SEMICONDUCTOR RECIPE

PMEffectAnalysis

Compare pre/post-PM populations, chamber behavior, traces and process stability using one shared context.

BEST FOR

measure PM effect · compare chamber before and after maintenance

AVOID WHEN

when pre/post windows or maintenance-event timing cannot be established

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[RCA Affected Control](#)[FDC Tool Chamber Compare](#)[SPC I-MR](#)[Box Distribution](#)[FDC Equipment Event Overlay](#)[FDC Chamber Fingerprint](#)

Reference Explorer

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RECIPE · SEMICONDUCTOR RECIPE

ChamberMatching

Compare chamber signatures and distributions to detect mismatch and identify driving variables.

BEST FOR

match chambers · compare chambers across a toolset

AVOID WHEN

when chambers do not share comparable process/recipe populations

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[FDC Chamber Fingerprint](#)[FDC Tool Chamber Compare](#)[FDC PCA Scores](#)[FDC PCA Loadings](#)[Ridge Distribution](#)[Wafer Comparison](#)

Reference Explorer

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RECIPE · SEMICONDUCTOR RECIPE

RcaCockpit

Run structured root-cause analysis with populations, commonality, contribution, evidence and genealogy in one cockpit.

BEST FOR

perform RCA · investigate root cause or commonality

AVOID WHEN

when the workflow needs causal proof from observational association alone

Advanced contract ·
› **configuration, data, accessibility, and authority**

Related capabilities

[RCA Affected Control](#)[RCA Commonality Ranking](#)[RCA Commonality Matrix](#)[RCA Contribution Waterfall](#)[RCA Evidence Matrix](#)[RCA Genealogy Graph](#)[RCA Fault Tree](#)[RCA Sankey](#)

Full Applications

Open a complete reference composition built from the department pattern and recipe authorities.

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SPC Control Center

A complete stability review composed from monitoring, dashboard, comparison and drill-down surfaces.

Is the critical-dimension process stable, and which observations need review?

critical dimension · control limits · review queue

Search records

Filter this application example

View

All records



critical dimension

control limits

review queue

Records

12

Full Applications

Open a complete reference composition built from the department pattern and recipe authorities.

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FDC Tool Health Center

A full operational surface for aligned traces, chamber context, event review and affected records.

Which tool or chamber signal changed around the observed equipment event?

pressure trace · tool/chamber · event review

Search records

Filter this application example

View

All records



pressure trace

tool/chamber

event review

Records

12

Full Applications

Open a complete reference composition built from the department pattern and recipe authorities.

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Excursion Investigation

An evidence-first investigation workspace for affected/control separation, commonality and supporting records.

What affected/control differences are worth investigating for this excursion?

affected vs control · evidence ledger · hypothesis review

Search records

Filter this application example

View

All records



affected vs control

evidence ledger

hypothesis review